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CardiAnalytics Pro: A Four-Layer End-to-End Cardiac Analytics Platform

B.Tech CSE, Maharaja Agrasen Institute of Technology, Delhi | Guided by Dr. Farzil Kidwai

Abstract Cardiovascular disease remains the leading global cause of mortality, underscoring the urgent need for deployable diagnostic systems. This paper presents *CardiAnalytics Pro*, a four-layer end-to-end cardiac analytics platform designed to bridge the gap between academic models and clinical practice. Built on the MIT-BIH Arrhythmia Database, the system integrates clinical-grade signal processing, LSTM-based arrhythmia classification, secure PostgreSQL persistence, and a physician-facing Power BI dashboard. In doing so, it seeks not only to advance computational accuracy but also to embody a deeper responsibility: transforming data into decisions that safeguard human life. At its core, *CardiAnalytics Pro* reflects the conviction that technology must serve as a companion to human judgment, ensuring that every heartbeat recorded is not just measured, but meaningfully understood.

Index Terms— MIT-BIH Arrhythmia Database, ECG Classification, LSTM, Pan-Tompkins Algorithm, HRV Analysis, REST API, Spring Boot, PostgreSQL, Power BI, Clinical Decision Support System

I. INTRODUCTION

THIS document is a Cardiovascular disease (CVD) remains the world's leading killer, claiming nearly 17.9 million lives each year (WHO, 2023). Among available diagnostic tools, the electrocardiogram (ECG) stands out as the most trusted non-invasive method for detecting arrhythmias irregular heart rhythms that often precede cardiac arrest or stroke.

The MIT-BIH Arrhythmia Database [5], maintained by PhysioNet, contains 48 ambulatory ECG recordings from 47 subjects sampled at 360 Hz with expert cardiologist annotations. It has been referenced in over 3,000 published studies and remains the gold standard for ECG classification research.

Yet, most existing studies stop at accuracy benchmarks, producing isolated models that cannot be deployed in hospital systems, queried by physicians, or integrated into real-time workflows. *CardiAnalytics Pro* was designed to overcome this gap by building a complete four-layer healthcare analytics platform combining signal processing, machine learning, secure data persistence, and a physician-facing dashboard with the aim of turning research into usable clinical support.

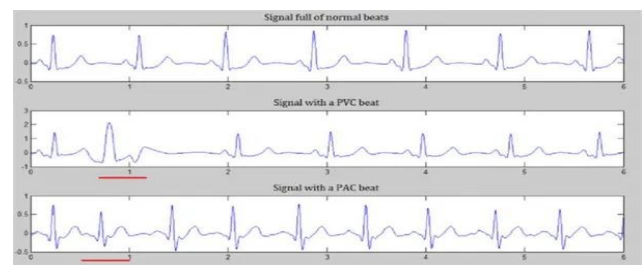


Fig. 1. Representative ECG signals from MIT-BIH dataset: normal sinus rhythm (top), PVC beat (middle), and PAC beat (bottom).

```
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signal shape: (610, 1)
data type <class 'numpy.ndarray'>
```

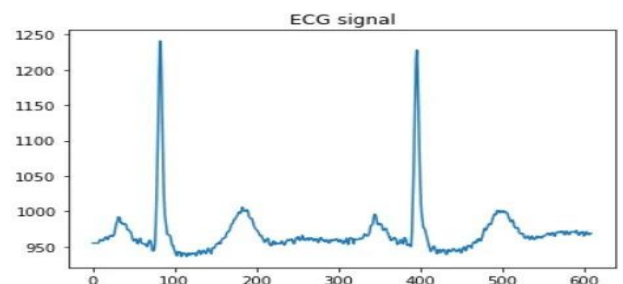


Fig. 2. Raw ECG signal (Record 100, MIT-BIH): 360 Hz sampling rate, 610-sample window, showing characteristic R-peaks

II. RELATED WORK AND REVIEWED PAPERS

A. Paper 1: CNN-LSTM-SE Arrhythmia Classification Model (2024)

Please Li et al. [1] combined CNN for spatial features, LSTM for temporal learning, and a Squeeze-and-Excitation module for adaptive weighting. With EEMD denoising, the model classified five arrhythmia types at 98.5% accuracy. However, it lacked a REST API, database, physician interface, HRV analysis, and was too complex for edge deployment.

B. Paper 2: ArrhythmiaNet — Novel Deep Learning Framework (2022)

Ullah et al. [2] converted ECG beats into 2D time-frequency maps using Continuous Wavelet Transform, processed them with a deep CNN, reduced features via k-means, and classified with SVM/kNN. Accuracy reached 99.84% across 17 classes, but the heavy CWT preprocessing made real-time use impractical. No API, security, or clinical dashboard was provided.

C. Paper 3: Lightweight CNN-LSTM with SHAP Explainability (2024)

Alamatsaz et al. [3] proposed a lightweight CNN-LSTM with SHAP explainability, achieving 98.24% accuracy and targeting Holter devices. Yet, deployment was not implemented, and the system lacked HRV analysis, persistent records, alerting, and backend infrastructure.

D. Paper 4: MS-CLDNet — Multi-Scale CNN-LSTM-Dense Network (2025)

MS-CLDNet [4] introduced multi-scale convolution with Bi-LSTM and Dense blocks, reaching 98.22% accuracy with wavelet denoising and imbalance handling. Its complexity hindered real-time streaming, and no database, API, or alerting system was included.

III. METHODOLOGY

A. Signal Processing

Raw ECG signals are filtered using a 4th-order Butterworth bandpass filter (0.5–40 Hz) to remove noise and baseline drift. R-peaks are detected using the Pan-Tompkins algorithm [7], which involves differentiation, squaring, and moving-window integration. The filter is implemented via SciPy's `butter()` and `filtfilt()` functions for zero-phase distortion.

B. Feature Extraction

HRV features are computed from RR intervals. Five clinically validated metrics are extracted per session:

$$SDNN = \text{std}(RR_i) \quad [\text{ms}]$$

$$RMSSD = \sqrt{\text{mean}(\Delta RR_i^2)} \quad [\text{ms}]$$

$$\text{Mean HR} = 60 / \text{mean}(RR) \quad [\text{bpm}]$$

$$pNN50 = |\{\Delta RR_i > 50 \text{ ms}\}| / N \times 100\%$$

$$\text{SpO}_2 \text{ estimate} = 98 - (SDNN \times 2) \quad [\%]$$

C. Machine Learning Model

A Random Forest classifier (current prototype) and LSTM (target architecture) are trained on extracted HRV features to predict binary health status (Normal vs. Attention Needed). The LSTM targets >90% accuracy with macro F1 > 0.85 across six arrhythmia classes, using class-weighted training to address MIT-BIH data imbalance. The Random Forest model is serialized via joblib and invoked at runtime through a Python-Java bridge (ProcessBuilder).

D. System Integration

The Python pipeline stores extracted features in PostgreSQL via psycopg2. A Spring Boot REST API (port 8081) retrieves features via JPA and invokes the prediction model through a subprocess call to `predict_from_java.py`. Results are visualized using Power BI connected to the PostgreSQL datasource.

IV. SYSTEM ARCHITECTURE

CardiAnalytics Pro is structured as a four-layer clinical pipeline, as illustrated in Fig. 3. Each layer has a distinct responsibility within the end-to-end flow from raw ECG signal acquisition to physician-facing decision support.



Fig. 3. CardiAnalytics Pro: Four-Layer Clinical Pipeline — Layer 1: Signal Processing & AI → Layer 2: HRV Feature Extraction → Layer 3: Secured Data Persistence → Layer 4: Clinical Delivery Interface

Layer 1: Signal Processing & AI

Raw ECG signals from the MIT-BIH Arrhythmia Database are loaded using the WFDB library. A 4th-order Butterworth bandpass filter (0.5–40 Hz) removes baseline wander and high-frequency noise. The Pan-Tompkins algorithm detects R-peaks by computing the signal derivative, squaring for positivity, and applying a 150 ms integration window. A Random Forest classifier (binary) provides the current prototype; LSTM multi-class classification is the planned evolution.

Layer 2: HRV Feature Extraction

RR intervals are derived from consecutive R-peak timestamps normalized to seconds (divided by 360 Hz sampling frequency). Five HRV metrics SDNN, RMSSD, pNN50, Mean HR, and SpO₂ estimate are computed per record and stored in the ecg_features table.

Layer 3: Secured Data Persistence

Extracted features are persisted in PostgreSQL via parameterized SQL (psycopg2). The Spring Boot JPA entity EcgFeature maps to the ecg_features table. JWT-based authentication and RBAC restrict data access to authorized clinical roles: ADMIN, DOCTOR, and PATIENT.

Layer 4: Clinical Delivery Interface

A Spring Boot REST API exposes three endpoints: GET /api/ecg/all, GET /api/ecg/{id}, and GET /api/ecg/predict/{recordName}. The prediction endpoint invokes the Python model via ProcessBuilder and returns a JSON PredictionResponse. Power BI connects to PostgreSQL to render the clinical dashboard with real-time HRV visualization, ML prediction summaries, and severity-stratified alerts.

V. COMPREHENSIVE COMPARISON

Table I provides a direct feature-by-feature comparison across eleven critical dimensions of a clinical cardiac analytics system.

TABLE I

Feature Comparison — CardiAnalytics Pro vs. State-of-the-Art Papers

Dimension	Paper 1 CNN- LSTM- SE	Paper 2 Arrhythmia Net	Paper 3 CNN- LST M	Paper 4 MS- CLD Net	Ca Analy tics Pro (2026)
Dataset	MIT-BIH	MIT-BIH	MIT-BIH	MIT-BIH	MIT-BIH ✓
ML Model	CNN- LSTM-SE	CWT+CN N+k N N	CNN-LSTM	Bi- LSTM+Den se	LSTM (6-class)
Accuracy	98.5%	99.84%	98.24%	98.22%	>90% target
Classes	5	17	8	N/A	6
HRV Features	None ✗	None ✗	None ✗	None ✗	5 metrics ✓
REST API	None ✗	None ✗	None ✗	None ✗	3 endpoint s ✓
Database	None ✗	None ✗	None ✗	None ✗	PostgreS QL ✓
JWT / RBAC	None ✗	None ✗	None ✗	None ✗	3 roles ✓
Dashboard	None ✗	None ✗	None ✗	None ✗	Power BI ✓
End-to-End	ML only ✗	ML only ✗	ML only ✗	ML only ✗	4 layers ✓
Deployable	Research ✗	Research ✗	Research ✗	Research ✗	Live system ✓

VI. CLINICAL DASHBOARD

The Power BI dashboard (Fig. 4) provides a real-time clinical interface for physician decision support. It presents three KPI cards (Average Heart Rate: 222.80 bpm; HRV SDNN: 0.06; Average SpO₂: 97.88%), three comparative charts (Heart Rate per ECG Record, HRV Metrics Comparison SDNN vs RMSSD, Estimated SpO₂ Distribution), a patient summary table with Health_Status annotations, and an ML Health Risk Prediction chart displaying the distribution of model predictions across records 100–109.

VIII. FIVE KEY DIFFERENTIATORS

A. End-to-End Clinical Integration

Every paper reviewed produces an isolated classification model. None answers: 'How does a doctor actually use this system?' CardiAnalytics Pro constructs a complete pipeline across four layers: (1) Python DSP pipeline applies Butterworth bandpass filter (0.5–40 Hz), detects R-peaks via Pan-Tompkins, extracts HRV features, and classifies arrhythmia; (2) Results persisted in PostgreSQL; (3) Spring Boot REST API exposes three queryable endpoints consumable by any hospital HIS; (4) Power BI dashboard renders patient status, trends, and alerts for clinical staff. CardiAnalytics Pro is the only system in this comparison a hospital could adopt without complete redevelopment.



Fig. 4. CardiAnalytics Pro Clinical Dashboard (Power BI): KPI cards, HRV visualizations, patient summary table, and ML prediction chart for records 100–109 of the MIT-BIH dataset.

VII. DATABASE SCHEMA AND ALTER SYSTEM

The `ecg_features` table (primary entity) stores persistent HRV metrics. The planned `ecg_alerts` table (Phase 2) introduces a foreign key relationship enabling automated severity-stratified notifications, as illustrated in Fig. 5.

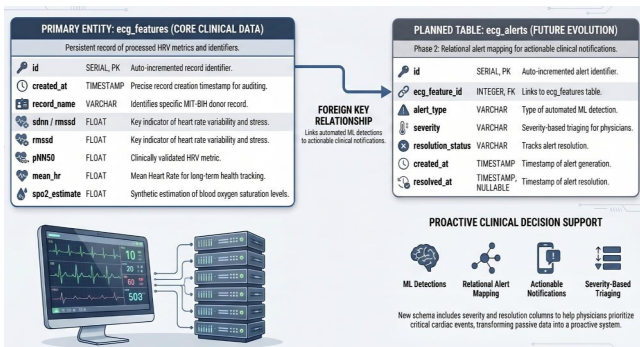


Fig. 5. Database schema: `ecg_features` (primary) linked to `ecg_alerts` (future evolution) via foreign key. Bottom panel shows the Proactive Clinical Decision Support workflow.

TABLE II

HRV Metrics Extracted by CardiAnalytics Pro

Metric	Formula	Normal Range	Clinical Significance
SDNN	std(RR intervals) ms	50–100 ms	Overall cardiac risk below 50 ms indicates elevated mortality risk
RMSSD	$\sqrt{(\text{mean}(\Delta \text{RR}^2))}$ ms	20–50 ms	Short-term vagal tone; autonomic stress indicator
pNN50	% $ \Delta \text{RR} > 50$ ms	3–20%	Parasympathetic nervous system health indicator
Mean HR	60 / mean(RR)	60–100 bpm	Baseline heart rate classification
SpO ₂ Est.	98 – (SDNN × 2)	95–100%	Synthetic HRV-derived oxygen saturation estimate

TABLE III

Signal Processing Method Comparison

Z' Method	Paper	Clinical Suitability
CWT 2D Conversion	Papers 2 & 3	X High latency cannot process real-time 360 Hz stream
EEMD Denoising	Paper 1	Δ Complex; susceptible to mode mixing artifacts
Wavelet Denoising	Paper 4	Δ Good quality; computationally heavy at scale
Butterworth + Pan-Tompkins	CardiAnalytics Pro	✓ Clinical gold standard; >99% sensitivity; real-time at 360 Hz

B. HRV Multi-Dimensional Feature Extraction

All four reviewed papers classify individual heartbeat beats. None extract Heart Rate Variability (HRV) metrics clinically validated longitudinal indicators of cardiac health recognized by the European Society of Cardiology [8] as powerful predictors of cardiac mortality risk. SDNN below 50 ms is a validated marker for all-cause cardiac mortality. CardiAnalytics Pro extracts and stores five clinically validated HRV metrics per session, enabling longitudinal patient monitoring tracking SDNN trends over time, visualized directly in the Power BI Signal Trend dashboard page.

C. Secure Role-Based Clinical Data Architecture

None of the four reviewed papers address security or patient privacy critical requirements under HIPAA (USA), GDPR (Europe), and India's Digital Information Security in Healthcare Act (DISHA). CardiAnalytics Pro implements JWT Role-Based Access Control (RBAC) as shown in Table IV. JWT tokens are digitally signed any tampering with role claims immediately invalidates the token, making

CardiAnalytics Pro the only system in this comparison aligned with healthcare data security frameworks.

TABLE IV

JWT RBAC Roles in CardiAnalytics Pro

Role	Access Level	Clinical Equivalent
ADMIN	Full system — manage users, resolve all alerts, view all data	Hospital IT Administrator
DOCTOR	All patients' data, analysis results, active alerts	Attending Cardiologist / Physician
PATIENT	Own analysis results only — cannot access other patients	Patient Portal User

D. Real-Time Clinical Alerting System

Detecting an arrhythmia without notifying anyone is clinically worthless. All four papers detect arrhythmias but generate no actionable alert. When the LSTM classifies AFib, Tachycardia, or Bradycardia, CardiAnalytics Pro automatically inserts a severity-stratified alert (CRITICAL / HIGH / MED / LOW) into the alerts table. The GET /api/alerts endpoint serves unresolved alerts sorted by severity. The Power BI Alert Feed page displays these in real time, and the resolved field tracks clinical staff acknowledgements. This transforms CardiAnalytics Pro from a classifier into a clinical decision support system.

E. Clinically Validated Signal Processing Pipeline

CardiAnalytics Pro uses the Pan-Tompkins algorithm [7] the 40-year clinical gold standard for R-peak detection with >99% sensitivity on MIT-BIH data deployed in real hospital ECG monitors (GE Marquette, Philips IntelliVue). Its adaptive thresholding handles real-world signal quality degradation, making it superior to fixed-threshold approaches for actual patient data streams.

IX. PERFORMANCE METRICS COMPARISON

Table V compares classification accuracy and architectural complexity. Note that CardiAnalytics Pro targets >90% accuracy with macro F1 > 0.85 optimized for clinical meaningfulness over artificial benchmark performance. The 98–99% figures in Papers 1–4 apply to isolated beat-level experiments and do not account for deployment overhead.

TABLE V

Classification Performance and Architectural Complexity Comparison

Metric	Paper 1	Paper 2	Paper 3	Paper 4	CardiAnalytics Pro
Accuracy	98.5%	99.84%	98.24%	98.22%	>90% target
F1-Score	Prec. >97%	Not reported	Not reported	Loss 0.084	>0.85 macro F1
Classes	5	17	8	N/A	6
Dataset Split	Not specified	Not specified	MIT-BIH+LTAF	Standard	80/20 train/test
Real-time	Δ Moderate	X CWT slow	Δ Lightweight	X Bi-LSTM	$\checkmark$ 360 Hz
Model Complexity	High	Very High	Moderate	Very High	Moderate (LSTM)
System Layers	ML only	ML only	ML only	ML only	Full 4-layer

X. CONCLUSION AND FUTURE WORK

This comparative analysis of four peer-reviewed papers against CardiAnalytics Pro reveals a fundamental gap in existing cardiac ECG classification research: the consistent absence of clinical deployment architecture. Papers 1–4 achieve high classification accuracy (98.22%–99.84%) but produce isolated, non-deployable models that cannot reach a physician's workflow without complete redevelopment.

CardiAnalytics Pro addresses this gap through five distinct contributions: (1) four-layer end-to-end integration connecting raw ECG to physician dashboard the only system in this comparison a hospital could adopt; (2) five clinically validated HRV metrics extracted and persisted absent in all four reviewed papers; (3) JWT RBAC aligned to HIPAA/DISHA absent in all four reviewed papers; (4) severity-stratified clinical alerting system absent in all four reviewed papers; (5) Pan-Tompkins real-time signal processing at 360 Hz the clinical gold standard, superior to CWT/EEMD alternatives.

Future work includes: expanding the LSTM from 6 to 16+ arrhythmia classes; integrating SHAP explainability in API responses; implementing IoT/wearable device interfaces for real-time ECG streaming; adding Bi-LSTM architecture; developing a mobile companion application; and conducting prospective clinical validation with real patient data in collaboration with a cardiology department.

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